

MSS - module 2

Written exam 2025/06/23

Table of Contents

MSS - module 2.....	1
Written exam 2025/06/23.....	1
Numerical part.....	1
Exercise 1.....	1
Exercise 2.....	4

Numerical part

Exercise 1

Consider the following DAE system.

$$\begin{cases} x'(t) - z'(t) = t^2 \\ x(t)y(t) = 1 \\ x(t)y(t)z(t) = t \end{cases}$$

rewrite in the form

$$\mathbf{E}(\mathbf{x}, t)\mathbf{x}' = \mathbf{G}(\mathbf{x}, t)$$

perform index reduction

$$\mathbf{E}^{(1)}(\mathbf{x}, t)\mathbf{x}' = \mathbf{G}^{(1)}(\mathbf{x}, t), \quad \dots \quad \mathbf{E}^{(k)}(\mathbf{x}, t)\mathbf{x}' = \mathbf{G}^{(k)}(\mathbf{x}, t),$$

until $\mathbf{E}^{(k)}(\mathbf{x}, t)$ is non singular and DAE is reduced to an ODE. Compute:

Compute the

1. differential index
2. Invariants at each index reduction step
3. Matrices $\mathbf{E}^{(j)}(\mathbf{x}, t)$ and vector $\mathbf{G}^{(j)}(\mathbf{x}, t)$ at each step

```
clear all

syms t x(t) y(t) z(t)

%Original system

vars = [x(t) y(t) z(t)];
DAE0 = [diff(x(t),t)-diff(z(t),t)-t^2;
        1-x(t)*y(t);
        t-x(t)*y(t)*z(t)]
```

DAE0 =

$$\begin{pmatrix} \frac{\partial}{\partial t} x(t) - \frac{\partial}{\partial t} z(t) - t^2 \\ 1 - x(t) y(t) \\ t - x(t) y(t) z(t) \end{pmatrix}$$

Step 0: Semi-explicit form: matrices E and G and invariants

```
[E0,G0] = massMatrixForm(DAE0,vars)
```

E0 =

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

G0 =

$$\begin{pmatrix} t^2 \\ x(t) y(t) - 1 \\ x(t) y(t) z(t) - t \end{pmatrix}$$

```
% Rank of E0. It is 1 < 3 thus is a DAE
rank(E0)
```

```
ans =
1
```

```
%Invariants
inv0 =G0(2:3)
```

inv0 =

$$\begin{pmatrix} x(t) y(t) - 1 \\ x(t) y(t) z(t) - t \end{pmatrix}$$

Step 1: Differentiate invariants and assemble the system

```
% Differentiation of invariants
inv0_diff =diff(inv0,t)
```

inv0_diff =

$$\begin{pmatrix} x(t) \frac{\partial}{\partial t} y(t) + y(t) \frac{\partial}{\partial t} x(t) \\ x(t) y(t) \frac{\partial}{\partial t} z(t) + x(t) z(t) \frac{\partial}{\partial t} y(t) + y(t) z(t) \frac{\partial}{\partial t} x(t) - 1 \end{pmatrix}$$

```
% DAE after 1st differentiation
DAE1 = E0*diff(vars).'-[G0(1);inv0_diff(1);inv0_diff(2)]
```

DAE1 =

$$\begin{pmatrix} \frac{\partial}{\partial t} x(t) - \frac{\partial}{\partial t} z(t) - t^2 \\ -x(t) \frac{\partial}{\partial t} y(t) - y(t) \frac{\partial}{\partial t} x(t) \\ 1 - x(t) z(t) \frac{\partial}{\partial t} y(t) - y(t) z(t) \frac{\partial}{\partial t} x(t) - x(t) y(t) \frac{\partial}{\partial t} z(t) \end{pmatrix}$$

%Explicit form

```
[E1,G1] = massMatrixForm(DAE1,vars)
```

E1 =

$$\begin{pmatrix} 1 & 0 & -1 \\ -y(t) & -x(t) & 0 \\ -y(t) z(t) & -x(t) z(t) & -x(t) y(t) \end{pmatrix}$$

G1 =

$$\begin{pmatrix} t^2 \\ 0 \\ -1 \end{pmatrix}$$

Perform LU decomposition (Triangularise E1)

% It is: L*U=E1

```
[L, U, p] = lu(E1)
```

L =

$$\begin{pmatrix} 1 & 0 & 0 \\ -y(t) & 1 & 0 \\ -y(t) z(t) & z(t) & 1 \end{pmatrix}$$

U =

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & -x(t) & -y(t) \\ 0 & 0 & -x(t) y(t) \end{pmatrix}$$

p = 3x3

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

% Compute E and G matrices with L matrix

```
E2 = L^(-1)*E1
```

E2 =

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & -x(t) & -y(t) \\ 0 & 0 & -x(t) y(t) \end{pmatrix}$$

```
G2 = L^(-1)*G1
```

G2 =

$$\begin{pmatrix} t^2 \\ t^2 y(t) \\ -1 \end{pmatrix}$$

```
% compute rank of (E2): 3 = 3 is a ODE
rank(E2)
```

```
ans =
3
```

```
ODE2 = E2*diff(vars) - G2
```

ODE2 =

$$\begin{pmatrix} \frac{\partial}{\partial t} x(t) - \frac{\partial}{\partial t} z(t) - t^2 \\ -x(t) \frac{\partial}{\partial t} y(t) - y(t) \frac{\partial}{\partial t} z(t) - t^2 y(t) \\ 1 - x(t) y(t) \frac{\partial}{\partial t} z(t) \end{pmatrix}$$

Exercise 2

Consider the following DAE system:

$$\begin{cases} x''(t) = \lambda(t)y(t) + \sin(t) \\ y''(t) = \lambda(t)x(t) + \cos(t) \\ y(t) = x(t)^2 - 1 \end{cases}$$

reduce to order 1 then using dummy variable reduce the DAE to index 1.

Compute the

1. differential index
2. DAE expressed with dummy variables
3. Invariants at each reduction step
4. DAE reduced to index 1
5. Projection to invariants method

```
clear all
```

```
syms t x(t) y(t) lambda(t)
% dummy variables
syms xdot(t) ydot(t)
```

```
%Transform original system with dummy variables
```

```
vars = [xdot(t), ydot(t) x(t) y(t) lambda(t)];
DAE0 = [diff(xdot(t),t)==lambda(t)*y(t)+sin(t);
        diff(ydot(t),t)==lambda(t)*x(t)+cos(t);
        diff(x(t),t)==xdot(t);
        diff(y(t),t)==ydot(t);
        y(t) == x(t)^2-1]
```

DAE0 =

$$\begin{pmatrix} \frac{\partial}{\partial t} xdot(t) = \sin(t) + \lambda(t) y(t) \\ \frac{\partial}{\partial t} ydot(t) = \cos(t) + \lambda(t) x(t) \\ \frac{\partial}{\partial t} x(t) = xdot(t) \\ \frac{\partial}{\partial t} y(t) = ydot(t) \\ y(t) = x(t)^2 - 1 \end{pmatrix}$$

Step 0: Semi-explicit form: matrices E and G and invariants

```
[E00,G00] = massMatrixForm(DAE0,vars)
```

E00 =

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

G00 =

$$\begin{pmatrix} \sin(t) + \lambda(t) y(t) \\ \cos(t) + \lambda(t) x(t) \\ xdot(t) \\ ydot(t) \\ x(t)^2 - y(t) - 1 \end{pmatrix}$$

```
% rank is 4 < 5 => system is DAE
rank(E00)
```

```
ans =
4
```

Compute the LU decomposition

```
[L00, U00, p00] = lu(E00)
```

L00 =

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

U00 =

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

p00 = 5x5

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

```
% L0*U0*p0 = E0
```

```
% p0^(-1)*L0^(-1)*E0 = U0
```

```
E0 = U00; G0 = p00^(-1)*L00^(-1)*G00
```

G0 =

$$\begin{pmatrix} \sin(t) + \lambda(t) y(t) \\ \cos(t) + \lambda(t) x(t) \\ \dot{x}(t) \\ \dot{y}(t) \\ x(t)^2 - y(t) - 1 \end{pmatrix}$$

```
E0*diff(vars,t) . '==G0
```

ans =

$$\begin{pmatrix} \frac{\partial}{\partial t} \dot{x}(t) = \sin(t) + \lambda(t) y(t) \\ \frac{\partial}{\partial t} \dot{y}(t) = \cos(t) + \lambda(t) x(t) \\ \frac{\partial}{\partial t} x(t) = \dot{x}(t) \\ \frac{\partial}{\partial t} y(t) = \dot{y}(t) \\ 0 = x(t)^2 - y(t) - 1 \end{pmatrix}$$

```
% F1st invariant
```

```
inv0 = G0(end)
```

```
inv0 = x(t)^2 - y(t) - 1
```

Step 1: Differentiate invariants and assemble the system

```
% Differentiation of invariants
```

```
inv0_diff =diff(inv0,t)
```

```
inv0_diff =
```

$$2 x(t) \frac{\partial}{\partial t} x(t) - \frac{\partial}{\partial t} y(t)$$

```
% DAE after 1st differentiation
```

```
DAE1 = E0*diff(vars).'-[G0(1:end-1);inv0_diff(1)]
```

```
DAE1 =
```

$$\begin{pmatrix} \frac{\partial}{\partial t} \text{xdot}(t) - \sin(t) - \lambda(t) y(t) \\ \frac{\partial}{\partial t} \text{ydot}(t) - \cos(t) - \lambda(t) x(t) \\ \frac{\partial}{\partial t} x(t) - \text{xdot}(t) \\ \frac{\partial}{\partial t} y(t) - \text{ydot}(t) \\ \frac{\partial}{\partial t} y(t) - 2 x(t) \frac{\partial}{\partial t} x(t) \end{pmatrix}$$

Compute E and G matrices step 1 after first differentiation

```
[E10,G10] = massMatrixForm(DAE1,vars)
```

```
E10 =
```

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2 x(t) & 1 & 0 \end{pmatrix}$$

```
G10 =
```

$$\begin{pmatrix} \sin(t) + \lambda(t) y(t) \\ \cos(t) + \lambda(t) x(t) \\ \text{xdot}(t) \\ \text{ydot}(t) \\ 0 \end{pmatrix}$$

```
%Rank is still < 5 => DAE
```

```
rank(E10)
```

```
ans =
```

```
4
```

Compute LU decomposition to

```
[L1, U1, p1] = lu(E10)
```

```
L1 =
```

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2x(t) & 1 & 1 \end{pmatrix}$$

U1 =

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

p1 = 5x5

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

E1 = U1; G1 = p1⁽⁻¹⁾*L1⁽⁻¹⁾*G10

G1 =

$$\begin{pmatrix} \sin(t) + \lambda(t) y(t) \\ \cos(t) + \lambda(t) x(t) \\ \dot{x}(t) \\ \dot{y}(t) \\ 2x(t)\dot{x}(t) - \dot{y}(t) \end{pmatrix}$$

E1*diff(vars,t).'==G1

ans =

$$\begin{pmatrix} \frac{\partial}{\partial t} \dot{x}(t) = \sin(t) + \lambda(t) y(t) \\ \frac{\partial}{\partial t} \dot{y}(t) = \cos(t) + \lambda(t) x(t) \\ \frac{\partial}{\partial t} x(t) = \dot{x}(t) \\ \frac{\partial}{\partial t} y(t) = \dot{y}(t) \\ 0 = 2x(t)\dot{x}(t) - \dot{y}(t) \end{pmatrix}$$

% Second invariant

inv1 = G1(end)

inv1 = 2x(t)xdot(t) - ydot(t)

Step 2: Differentiate invariants and assemble the system

% Differentiation of invariants


```
inv1_diff =diff(inv1,t)
```

```
inv1_diff =
```

$$2x(t)\frac{\partial}{\partial t} \dot{x}(t) + 2\dot{x}(t)\frac{\partial}{\partial t} x(t) - \frac{\partial}{\partial t} \dot{y}(t)$$

```
% DAE after 2nd differentiation
```

```
DAE2 = E1*diff(vars).'-[G1(1:end-1);inv1_diff(1)]
```

```
DAE2 =
```

$$\begin{pmatrix} \frac{\partial}{\partial t} \dot{x}(t) - \sin(t) - \lambda(t) y(t) \\ \frac{\partial}{\partial t} \dot{y}(t) - \cos(t) - \lambda(t) x(t) \\ \frac{\partial}{\partial t} x(t) - \dot{x}(t) \\ \frac{\partial}{\partial t} y(t) - \dot{y}(t) \\ \frac{\partial}{\partial t} \dot{y}(t) - 2\dot{x}(t)\frac{\partial}{\partial t} x(t) - 2x(t)\frac{\partial}{\partial t} \dot{x}(t) \end{pmatrix}$$

Compute E and G matrices step 2 after second differentiation

```
[E20,G20] = massMatrixForm(DAE2,vars)
```

```
E20 =
```

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -2x(t) & 1 & -2\dot{x}(t) & 0 & 0 \end{pmatrix}$$

```
G20 =
```

$$\begin{pmatrix} \sin(t) + \lambda(t) y(t) \\ \cos(t) + \lambda(t) x(t) \\ \dot{x}(t) \\ \dot{y}(t) \\ 0 \end{pmatrix}$$

```
rank(E20) % rank is still < 5 => system is still DAE
```

```
ans =
```

```
4
```

Compute LU decomposition to

```
[L2, U2, p2] = lu(E20)
```

```
L2 =
```

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -2x(t) & 1 & -2\dot{x}(t) & 0 & 1 \end{pmatrix}$$

U2 =

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

p2 = 5x5

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$E2 = U2; G2 = p2^{(-1)} * L2^{(-1)} * G20$$

G2 =

$$\begin{pmatrix} \sin(t) + \lambda(t) y(t) \\ \cos(t) + \lambda(t) x(t) \\ \dot{x}(t) \\ \dot{y}(t) \\ 2\dot{x}(t)^2 - \cos(t) - \lambda(t) x(t) + 2x(t) (\sin(t) + \lambda(t) y(t)) \end{pmatrix}$$

$$E2 * \text{diff}(\text{vars}, t) \cdot ' == G2$$

ans =

$$\begin{pmatrix} \frac{\partial}{\partial t} \dot{x}(t) = \sigma_1 \\ \frac{\partial}{\partial t} \dot{y}(t) = \cos(t) + \lambda(t) x(t) \\ \frac{\partial}{\partial t} x(t) = \dot{x}(t) \\ \frac{\partial}{\partial t} y(t) = \dot{y}(t) \\ 0 = 2\dot{x}(t)^2 - \cos(t) - \lambda(t) x(t) + 2x(t) \sigma_1 \end{pmatrix}$$

where

$$\sigma_1 = \sin(t) + \lambda(t) y(t)$$

Invariant 2 and isolate lambda(t)

$$\text{inv2} = G2(\text{end})$$

$$\text{inv2} = 2 \dot{x}(t)^2 - \cos(t) - \lambda(t) x(t) + 2 x(t) (\sin(t) + \lambda(t) y(t))$$

```
isolate(inv2 == 0, lambda(t))
```

ans =

$$\lambda(t) = \frac{2 \dot{x}(t)^2 - \cos(t) + 2 \sin(t) x(t)}{x(t) - 2 x(t) y(t)}$$

Step 3: Differentiate invariants and assemble the system

```
% Differentiation of invariants
```

```
inv2_diff = diff(inv2, t)
```

inv2_diff =

$$\sin(t) - \lambda(t) \frac{\partial}{\partial t} x(t) - x(t) \frac{\partial}{\partial t} \lambda(t) + 4 \dot{x}(t) \frac{\partial}{\partial t} \dot{x}(t) + 2 x(t) \left(\cos(t) + \lambda(t) \frac{\partial}{\partial t} y(t) + y(t) \frac{\partial}{\partial t} \lambda(t) \right) + 2 \dot{y}(t)$$

```
% DAE after 3rd differentiation
```

```
DAE3 = E2*diff(vars).'-[G2(1:end-1);inv2_diff(1)]
```

DAE3 =

$$\begin{pmatrix} \frac{\partial}{\partial t} \dot{x}(t) - \sin(t) - \sigma_1 \\ \frac{\partial}{\partial t} \dot{y}(t) - \cos(t) - \lambda(t) x(t) \\ \frac{\partial}{\partial t} x(t) - \dot{x}(t) \\ \frac{\partial}{\partial t} y(t) - \dot{y}(t) \\ \lambda(t) \frac{\partial}{\partial t} x(t) - \sin(t) + x(t) \frac{\partial}{\partial t} \lambda(t) - 4 \dot{x}(t) \frac{\partial}{\partial t} \dot{x}(t) - 2 x(t) \left(\cos(t) + \lambda(t) \frac{\partial}{\partial t} y(t) + y(t) \frac{\partial}{\partial t} \lambda(t) \right) - 2 \dot{y}(t) \end{pmatrix}$$

where

$$\sigma_1 = \lambda(t) y(t)$$

Compute E and G matrices step 3 after third differentiation

```
[E30,G30] = massMatrixForm(DAE3,vars)
```

E30 =

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -4 \dot{x}(t) & 0 & \lambda(t) - 2 \sin(t) - 2 \lambda(t) y(t) & -2 \lambda(t) x(t) & x(t) - 2 x(t) y(t) \end{pmatrix}$$

G30 =

$$\begin{pmatrix} \sin(t) + \lambda(t) y(t) \\ \cos(t) + \lambda(t) x(t) \\ \dot{x}(t) \\ \dot{y}(t) \\ \sin(t) + 2 \cos(t) x(t) \end{pmatrix}$$

Compute LU decomposition. U3 matrix is triangular and full rank. system is index 3

$$[L3, U3, p3] = \text{lu}(E30)$$

$$L3 =$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -4 \dot{x}(t) & 0 & \lambda(t) - 2 \sin(t) - 2 \lambda(t) y(t) & -2 \lambda(t) x(t) & 1 \end{pmatrix}$$

$$U3 =$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & x(t) - 2 x(t) y(t) \end{pmatrix}$$

$$p3 = 5 \times 5$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$